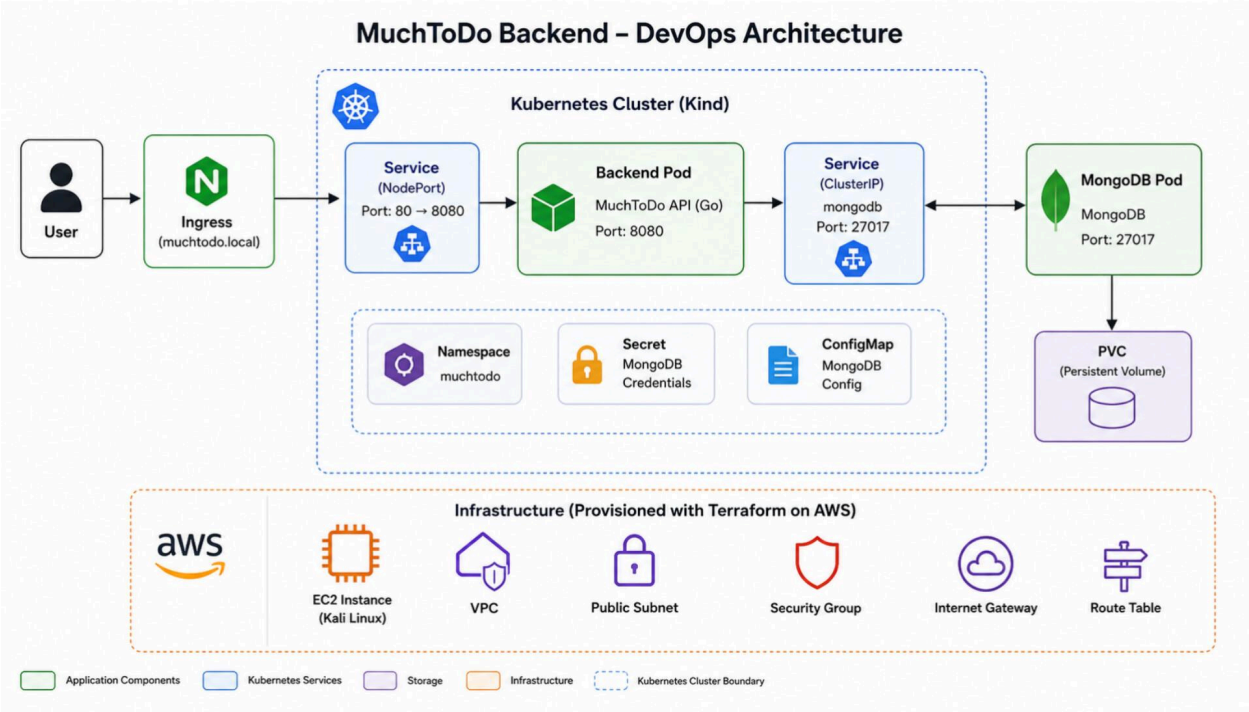
From Terraform to Kubernetes: Building and Deploying Backend Like a Real DevOps Engineer

Full End-to-End “MuchToDo” Project Documentation



Introduction

This project was a full hands-on DevOps assessment using a Backend API “MuchToDo”.

The goal was to move from infrastructure provisioning all the way to a working Kubernetes deployment using production-style best practices.

This included:

- AWS EC2 provisioning with Terraform modules
- Custom VPC and networking setup
- Docker installation and optimization
- Backend containerization
- Docker Compose services
- MongoDB persistent storage
- Kubernetes deployment using KIND
- Services and Ingress configuration
- Full API validation

This article documents exactly what was done, what failed, what was fixed, and what was learned.

PHASE 1 - Infrastructure Provisioning with Terraform

Objective

Provision a clean AWS environment instead of relying on default infrastructure.

What Was Built

Using Terraform modules:

- Custom VPC
- Public subnet
- Internet Gateway
- Route table
- Route table association
- Security Group

- Kali Linux EC2 instance
- Key pair access

Why This Matters

Senior engineers avoid unmanaged infrastructure.

Infrastructure as Code improves:

- repeatability
- scalability
- team collaboration
- disaster recovery
- production reliability

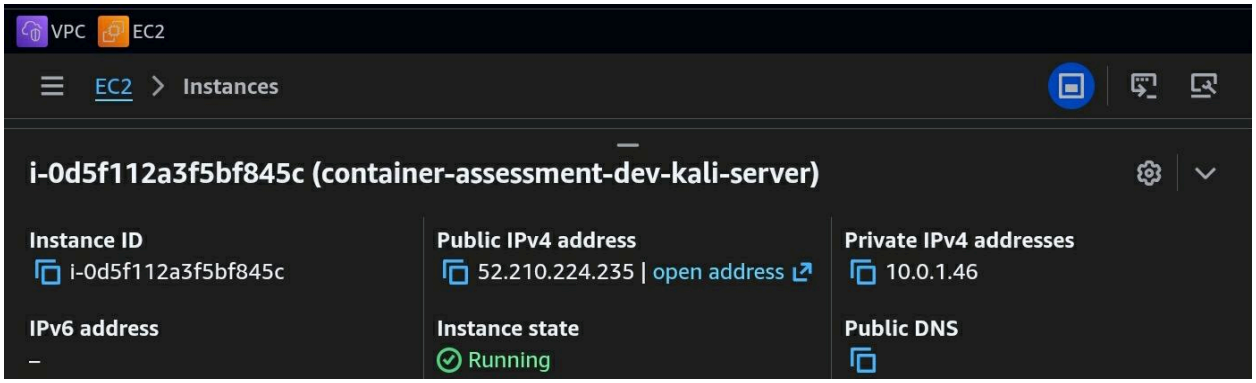
📸 Infra Setup main.tf

```
~/container-assessment/infra $ ls
main.tf     provider.tf               terraform.tfvars
modules     terraform.tfstate         variables.tf
outputs.tf  terraform.tfstate.backup
~/container-assessment/infra $ cat main.tf
module "vpc" {
  source = "./modules/vpc"

  project_name       = var.project_name
  environment        = var.environment
  vpc_cidr           = "10.0.0.0/16"
  public_subnet_cidr = "10.0.1.0/24"
  availability_zone  = "eu-west-1a"
}

module "security_group" {
  source = "./modules/security-group"

  project_name = var.project_name
  environment  = var.environment
  vpc_id       = module.vpc.vpc_id
}

module "ec2" {
  source = "./modules/ec2"

  project_name      = var.project_name
  environment       = var.environment
  ami_id            = var.ami_id
  instance_type     = var.instance_type
  key_name          = var.key_name
  subnet_id         = module.vpc.public_subnet_id
  security_group_id = module.security_group.security_group_id
}
~/container-assessment/infra $
```

📸 Public IP output from Terraform

```
module.vpc.aws_subnet.public: Creation complete after 12s [id=su
bnet-06a8f71e381fb0c93]
module.vpc.aws_route_table_association.public: Creating...
module.ec2.aws_instance.kali: Creating...
module.vpc.aws_route_table_association.public: Creation complete
 after 1s [id=rtbassoc-0a01cd7773af3a04f]
module.ec2.aws_instance.kali: Still creating... [00m10s elapsed]
module.ec2.aws_instance.kali: Creation complete after 15s [id=i-
0d5f112a3f5bf845c]

Apply complete! Resources: 8 added, 0 changed, 0 destroyed.

Outputs:

instance_id = "i-0d5f112a3f5bf845c"
instance_public_ip = "52.210.224.235"
ssh_command = "ssh -i ~/container-assessment/Adejare.pem kali@52
.210.224.235"
~/container-assessment/infra $
```

📸 EC2 instance created successfully



PHASE 2 - SSH Access and Server Validation

Objective

Connect to the EC2 instance and validate system readiness.

Validation Performed

Commands used:

- whoami
- uname -a
- df -h
- free -h

Key Discovery

The server had low memory for reliable Docker builds.

Fix Applied

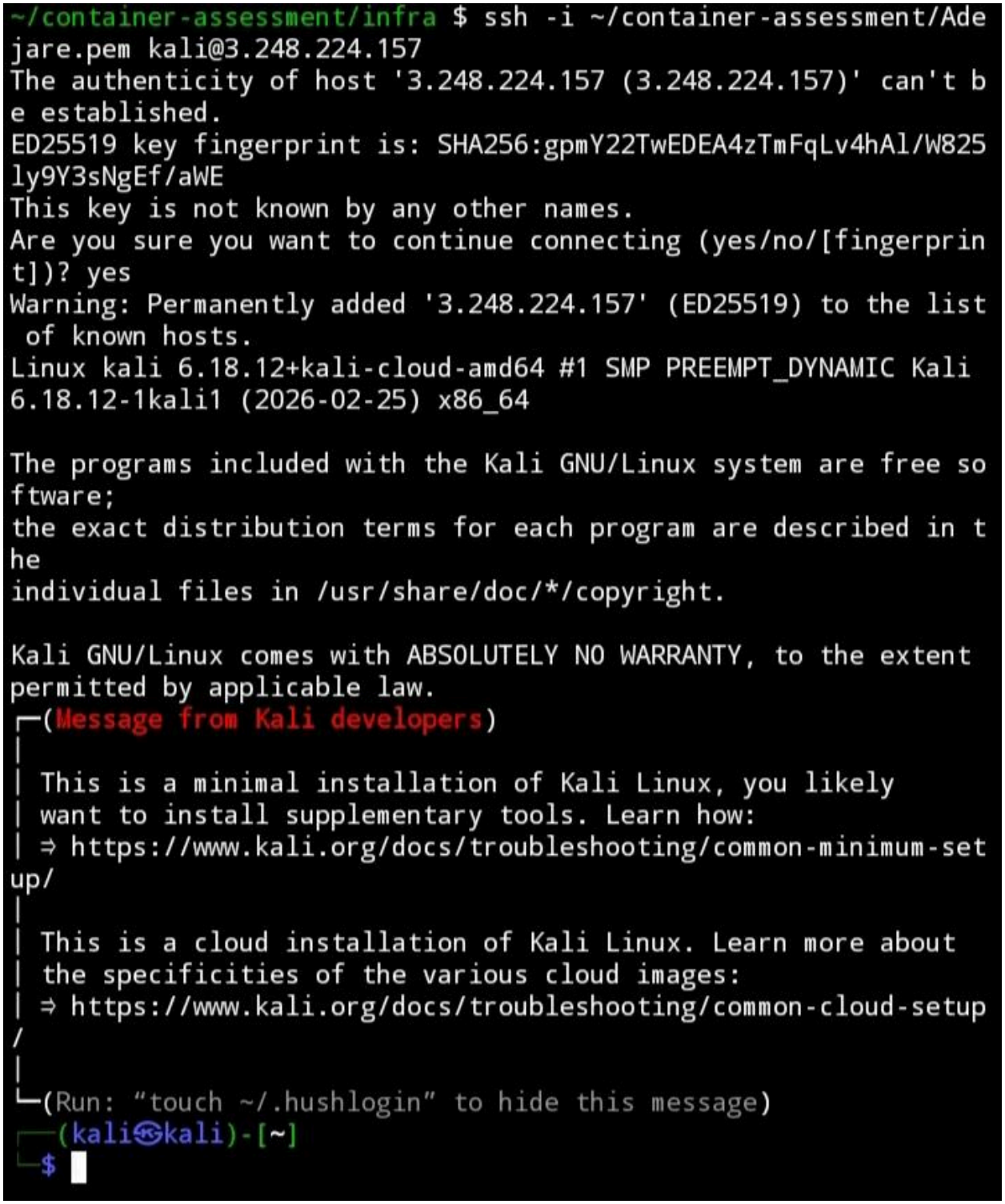
Added 2GB swap memory.

This prevented future Docker build crashes.

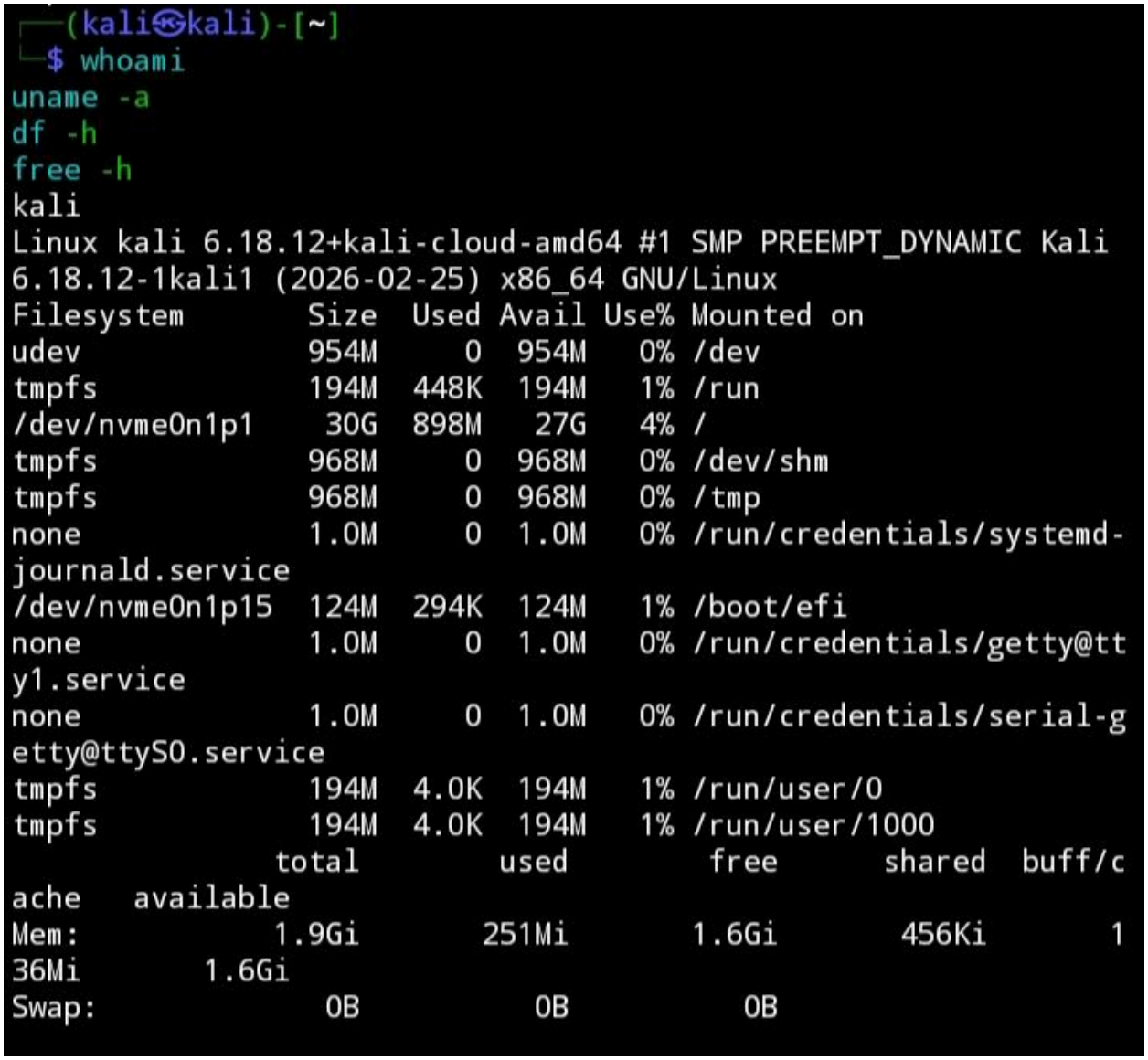
Lesson

Senior DevOps work starts with environment stability.

📸 SSH login successful



📸 Memory before swap



PHASE 3 - Server Dependencies Setups and Installation (Ansible)

```
└─$ cat setup.yml
---
- name: Provision Backend Environment (Kali Safe)
  hosts: localhost
  connection: local
  become: true


  vars:
    swap_file_size: 2G
    project_dir: /home/kali/much-to-do

  tasks:

    # ------------------------
    # FIX DOCKER REPO ISSUE (KALI SPECIFIC)
    # ------------------------
    - name: Remove broken Docker repo if present
      file:
        path: /etc/apt/sources.list.d/docker.list
        state: absent

    - name: Prevent Docker repo from being re-added
      copy:
        dest: /etc/apt/preferences.d/no-docker
        content: |
          Package: *
          Pin: origin download.docker.com
          Pin-Priority: -1

    - name: Update apt cache
      apt:
        update_cache: yes

    # ------------------------
    # INSTALL SYSTEM DEPENDENCIES
    # ------------------------
    - name: Install required packages
      apt:
        name:
          - docker.io
          - docker-compose
          - git
          - curl
        state: present

    # ------------------------
    # ENABLE DOCKER
```

What Ansible Playbook (setup.yml) Did -

1. Fixes Kali + Docker repository conflict (Kali-specific)

  - Deletes any broken /etc/apt/sources.list.d/docker.list file

  - Creates a permanent apt pin (/etc/apt/preferences.d/no-docker) that blocks the official Docker repo forever

  - Runs apt update

2. Installs system dependencies

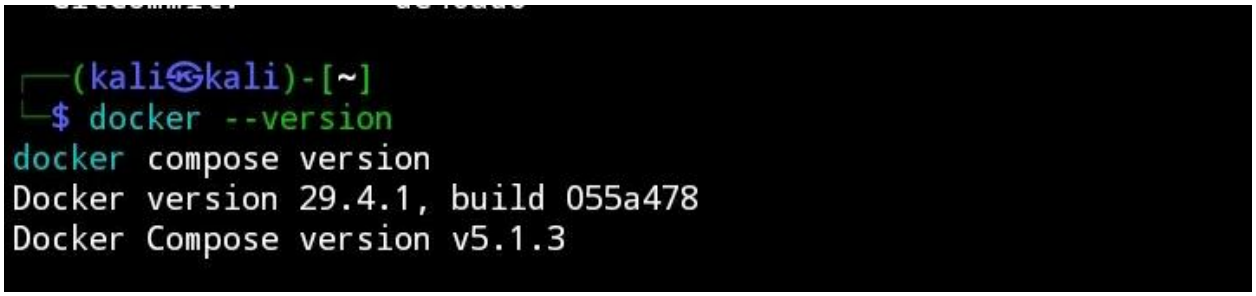
- docker.io (Debian-packaged version - safe on Kali)

- docker-compose

- git

- curl

📸 docker --version output



3. Enables Docker properly

 - Starts and enables the Docker service on boot

 - Adds the kali user to the docker group (so you can run docker commands without sudo)

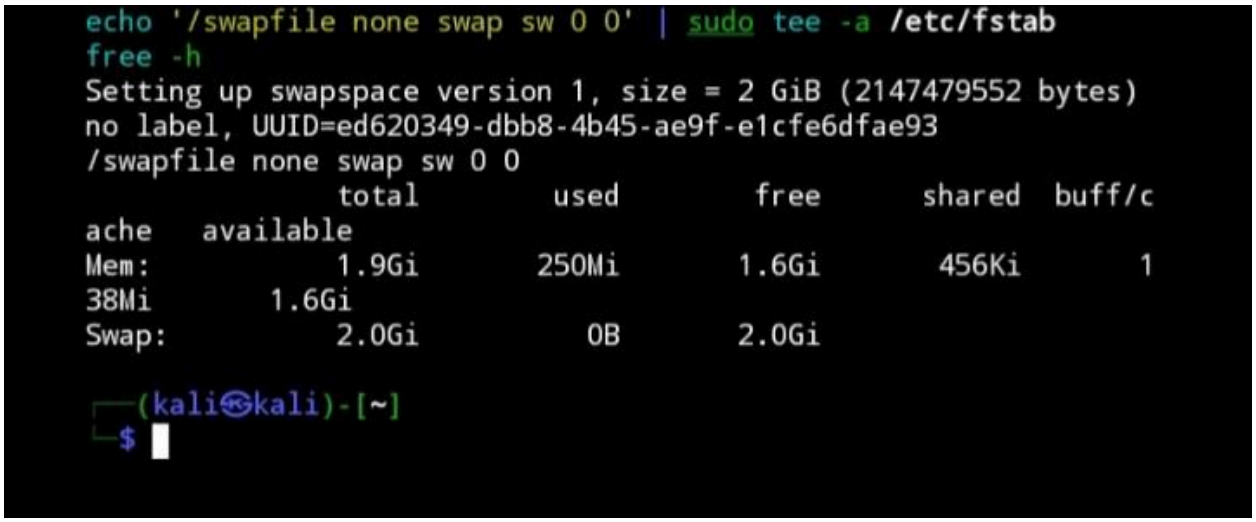
4. Creates 2 GB swap file (critical for builds)

 - Checks if /swapfile already exists

```
    enabled: yes
    state: started

- name: Add kali user to docker group
  user:
    name: kali
    groups: docker
    append: yes

# -------------------------
# SWAP (CRITICAL FOR BUILDS)
# -------------------------
- name: Check if swapfile exists
  stat:
    path: /swapfile
  register: swapfile

- name: Create swap file
  command: fallocate -l {{ swap_file_size }} /swapfile
  when: not swapfile.stat.exists

- name: Set swap permissions
  file:
    path: /swapfile
    mode: '0600'
  when: not swapfile.stat.exists

- name: Format swap
  command: mkswap /swapfile
  when: not swapfile.stat.exists

- name: Enable swap
  command: swapon /swapfile
  when: not swapfile.stat.exists

- name: Persist swap in fstab
  lineinfile:
    path: /etc/fstab
    line: "/swapfile none swap sw 0 0"
    state: present

# -------------------------
# PROJECT DIRECTORY SETUP
# -------------------------
- name: Ensure project directory exists
  file:
    path: "{{ project_dir }}"
    state: directory
```

- If not: creates a 2 GB swap file, sets secure permissions, formats it, enables it, and adds it to /etc/fstab so it survives reboots

📸 Memory after swap enabled



5. Sets up your project folder

- Creates the directory /home/kali/much-to-do

- Fixes ownership and permissions on the project folder and the entire /home/kali directory

---

PHASE 4 - Project Structure Inspection

Backend path:

~/Docker-k8-handson

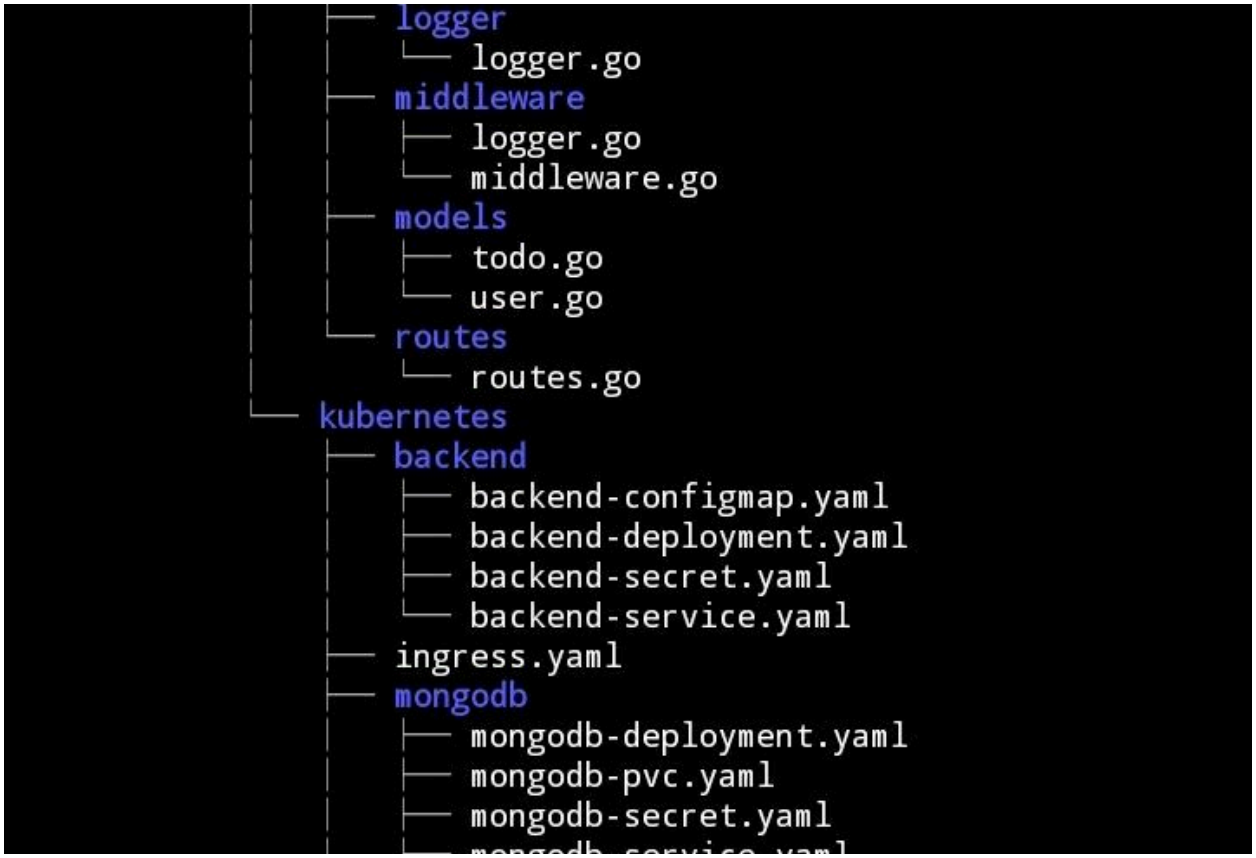
Why This Matters

Wrong Docker build context causes broken deployments.

Engineers inspect first.
They do not assume.

📸 Project structure view

```
┌──(kali㉿kali)-[~/Docker-k8-handson]
└─$ tree
.
├── ansible
│   └── setup.yml
├── infra
│   ├── main.tf
│   ├── modules
│   │   ├── ec2
│   │   │   ├── main.tf
│   │   │   ├── outputs.tf
│   │   │   └── variables.tf
│   │   ├── security-group
│   │   │   ├── main.tf
│   │   │   ├── output.tf
│   │   │   ├── outputs.tf
│   │   │   └── variables.tf
│   │   └── vpc
│   │       ├── main.tf
│   │       ├── outputs.tf
│   │       └── variables.tf
│   ├── outputs.tf
│   ├── provider.tf
│   └── variables.tf
└── much-to-do
    └── Server
        └── MuchToDo
            ├── Dockerfile
            ├── Makefile
            ├── cmd
            │   └── api
            │       └── main.go
            ├── docker-compose.yaml
            ├── docs
            │   ├── docs.go
            │   ├── swagger.json
            │   └── swagger.yaml
            ├── go.mod
            ├── go.sum
            ├── internal
            │   ├── auth
            │   │   ├── auth.go
            │   │   └── auth_test.go
            │   ├── cache
            │   │   └── cache.go
            │   ├── config
            │   │   └── config.go
```



PHASE 5 - Phase 1 – Application Analysis

Key Findings

- Located Backend Entry point at:

cmd/api/main.go

I ran

go run cmd/api/main.go

- Go Version Mismatch

Problem:

go.mod required Go 1.25.1 but Dockerfile used Go 1.22

Fix

Updated Dockerfile builder image to supported Go version.

```
└─$ cat Dockerfile
FROM golang:1.25-alpine AS builder

WORKDIR /app
RUN apk add --no-cache git

COPY go.mod go.sum ./
RUN go mod download

COPY . .

RUN CGO_ENABLED=0 GOOS=linux GOARCH=amd64 \
    go build -ldflags="-w -s" -o server ./cmd/api


FROM alpine:3.19

RUN apk add --no-cache ca-certificates
RUN adduser -D appuser

WORKDIR /app

COPY --from=builder /app/server .

USER appuser

EXPOSE 8080

CMD ["./server"]
```

What Dockerfile Does

- Multi-stage build reduces image size
- Static binary compilation

- Environmental Configuration issue

Could not connect to MongoDB
error parsing uri: scheme must be "mongodb"

Root Cause

- `MONGO_URI` was empty
- Config loader (Viper) depended on env variables

Fix

export MONGO_URI="mongodb://localhost:27017"

---

2 - Docker Compose Optimization

```
└─$ cat docker-compose.yaml
services:
  mongodb:
    image: mongo:7
    ports:
      - "27017:27017"
    environment:
      MONGO_INITDB_ROOT_USERNAME: root
      MONGO_INITDB_ROOT_PASSWORD: example

  backend:
    build: .
    ports:
      - "8080:8080"
    environment:
      MONGO_URI: mongodb://root:example@mongodb:27017/?authSourc
e=admin
    depends_on:
      - mongodb
```

docker compose up -d
curl http://localhost:8080/health

```
┌──(kali㉿kali)-[~]
└─$ docker ps
CONTAINER ID   IMAGE              COMMAND                  CREAT
ED         STATUS         PORTS
           NAMES
5223fc02ef05   muchtodo-backend   "./server"               9 min
utes ago   Up 9 minutes   0.0.0.0:8080->8080/tcp, [::]:8080->808
0/tcp      muchtodo-backend-1
68befbbed680   mongo:7            "docker-entrypoint.s…"   9 min
utes ago   Up 9 minutes   0.0.0.0:27017->27017/tcp, [::]:27017->
27017/tcp  muchtodo-mongodb-1

┌──(kali㉿kali)-[~]
└─$ curl http://localhost:8080/health
{"cache":"disabled","database":"ok"}
```

{"cache":"disabled","database":"ok"}

Phase 6 - Kubernetes Deployment with KIND

Backend Deployment

- Replica count: 2
- Container port: 8080
- Env vars injected

MongoDB Deployment

- Replica: 1
- Persistent Volume used

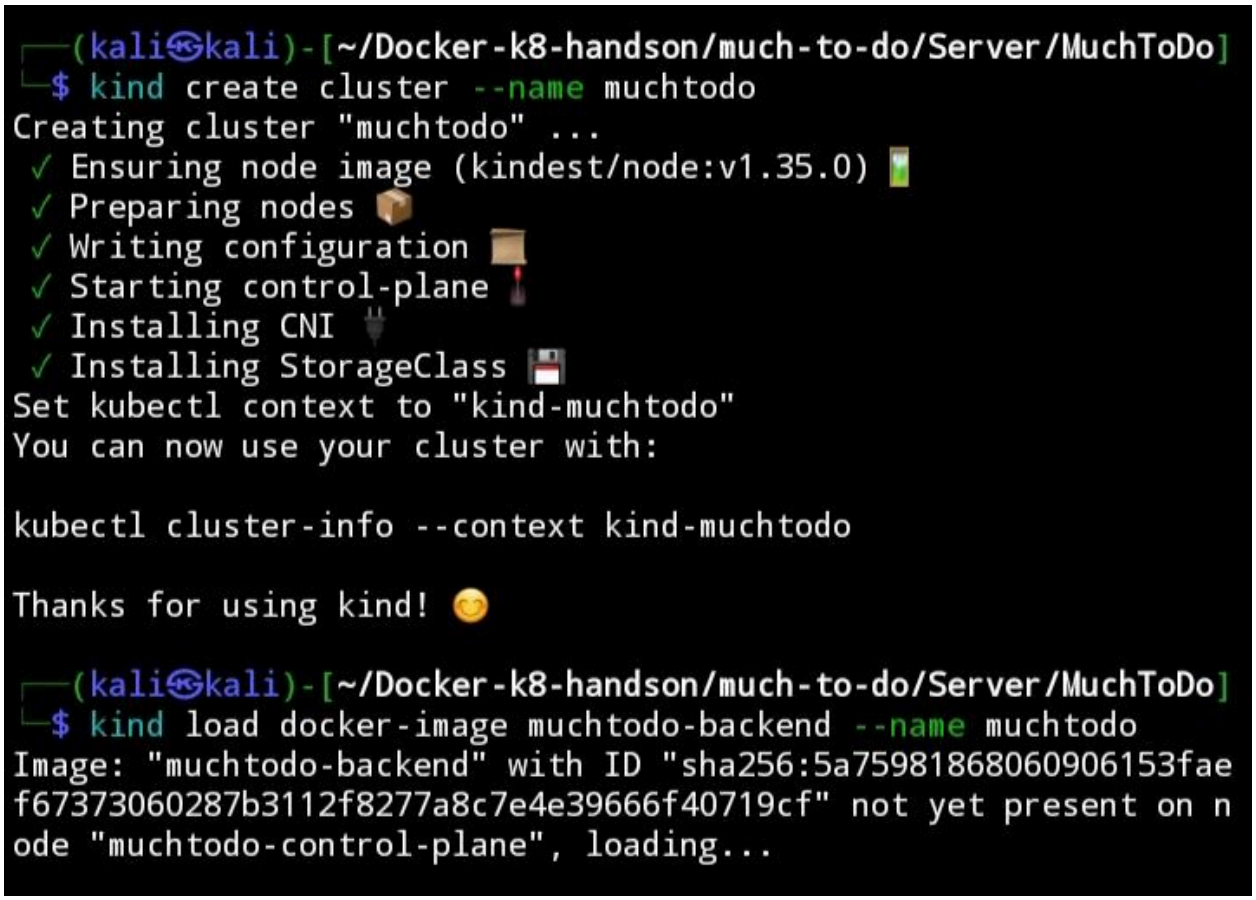
What Was Created

- Namespace
- Secret
- ConfigMap
- PVC

- MongoDB Deployment
- MongoDB Service
- Backend Deployment
- Backend Service
- Ingress Resource

Cluster Creation

kind create cluster --name muchtodo



…

…

```
┌──(kali㉿kali)-[~/much-to-do]
└─$ nano ~/much-to-do/kubernetes/mongodb/mongodb-secret.yaml

┌──(kali㉿kali)-[~/much-to-do]
└─$ kubectl apply -f kubernetes/mongodb/mongodb-secret.yaml
secret/mongodb-secret created

┌──(kali㉿kali)-[~/much-to-do]
└─$ kubectl get secrets -n muchtodo
NAME             TYPE     DATA   AGE
mongodb-secret   Opaque   2      19s

┌──(kali㉿kali)-[~/much-to-do]
└─$ nano ~/much-to-do/kubernetes/mongodb/mongodb-configmap.yaml

┌──(kali㉿kali)-[~/much-to-do]
└─$ kubectl apply -f kubernetes/mongodb/mongodb-configmap.yaml
configmap/mongodb-config created

┌──(kali㉿kali)-[~/much-to-do]
└─$ kubectl get configmaps -n muchtodo
NAME               DATA   AGE
kube-root-ca.crt   1      3m54s
mongodb-config     1      7s

┌──(kali㉿kali)-[~/much-to-do]
└─$ nano ~/much-to-do/kubernetes/mongodb/mongodb-pvc.yaml

┌──(kali㉿kali)-[~/much-to-do]
└─$ kubectl apply -f kubernetes/mongodb/mongodb-pvc.yaml
persistentvolumeclaim/mongodb-pvc created
```

📸 kubectl get nodes

```
└─$ kubectl get nodes

NAME                   STATUS   ROLES           AGE     VERSIO
N
muchtodo-control-plane   Ready    control-plane   5h34m   v1.35.
0
```

📸 kubectl get namespaces

```
└─$ kubectl get namespaces

NAME                 STATUS   AGE
default              Active   5h34m
ingress-nginx        Active   5h3m
kube-node-lease      Active   5h34m
kube-public          Active   5h34m
kube-system          Active   5h34m
local-path-storage   Active   5h34m
muchtodo             Active   5h22m
```

📸 kubectl get pvc -A

```
└─$ kubectl get pvc -A
NAMESPACE   NAME          STATUS   VOLUME
            CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATT
RIBUTESCLASS   AGE
muchtodo    mongodb-pvc   Bound    pvc-d6a9de76-e8ab-443f-a54b-4
c09cbd8fe67   1Gi        RWO            standard       <unset>
              5h19m
```

📸 kubectl get pods

```
┌──(kali㉿kali)-[~/Docker-k8-handson/much-to-do/Server/MuchToDo]
└─$ kubectl get pods -n muchtodo
NAME                       READY   STATUS    RESTARTS   AGE
backend-8b848967b-5dg77    1/1     Running   0          52s
backend-8b848967b-csvgv    1/1     Running   0          52s
mongodb-7c75b4b968-mvhq4   1/1     Running   0          12m
```

PHASE 7 - Ingress Configuration / Validation

Ingress YAML Logic

- Host: `muchtodo.local`
- Path: `/`
- Service: backend-service

Added Domain Mapping in /etc/hosts

Mapped 127.0.0.1 <<>> muchtodo.local

```
  GNU nano 8.7.1                  /etc/hosts
# Your system has configured 'manage_etc_hosts' as True.
# As a result, if you wish for changes to this file to persist
# then you will need to either
# a.) make changes to the master file in /etc/cloud/templates/h>
# b.) change or remove the value of 'manage_etc_hosts' in
#     /etc/cloud/cloud.cfg or cloud-config from user-data
#
127.0.1.1 ip-10-0-1-46.eu-west-1.compute.internal kali
127.0.0.1 muchtodo.local

# The following lines are desirable for IPv6 capable hosts
::1 localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

```
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters


┌──(kali㉿kali)-[~/Docker-k8-handson/much-to-do/Server/MuchToDo]
└─$ kubectl get ingress -n muchtodo
NAME              CLASS   HOSTS           ADDRESS     PORTS
AGE
muchtodo-ingress   nginx   muchtodo.local   localhost   80
5h22m
```

PHASE 8 - Ingress Errors and Fixes

Ingress Not Accessible

Symptoms

curl http://muchtodo.local/health

Error:

Could not connect to server

Root Cause Analysis

Layer	Status
Pods	OK
Services	OK
Ingress	OK
External Access	FAIL

Real Issue

Kind networking does NOT expose:

- LoadBalancer externally
- NodePort to localhost directly

Debugging Process

Step 1: Check ingress

kubectl get ingress -n muchtodo

Result:

ADDRESS: localhost

Step 2: Check ingress service

kubectl get svc -n ingress-nginx

Result:

LoadBalancer (EXTERNAL-IP: pending)

Get NodePort

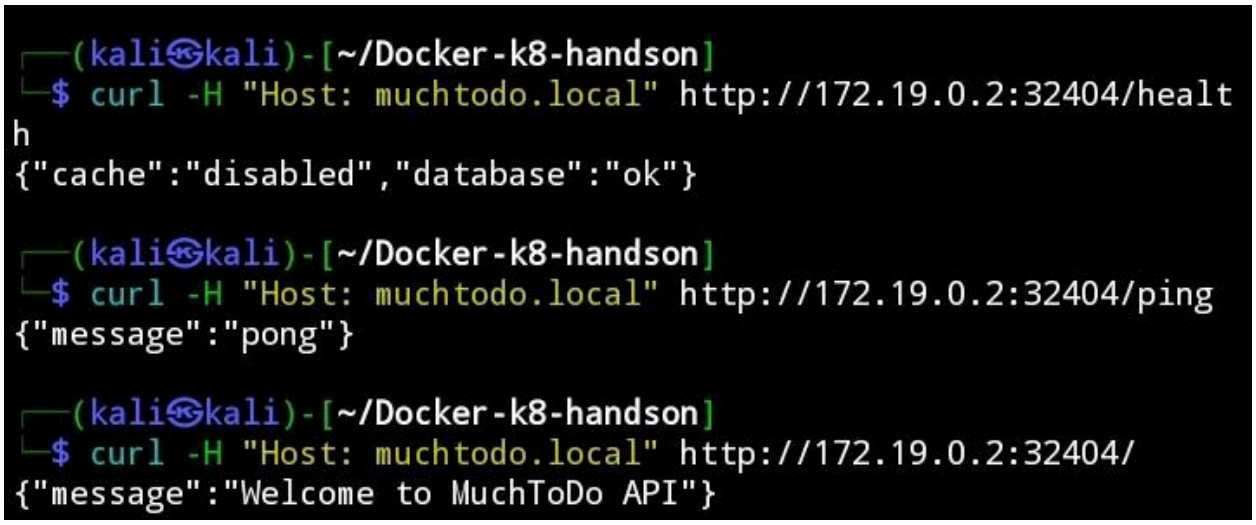
kubectl get svc -n ingress-nginx

Output:

80:32404/TCP

Final Working Request

curl -H "Host: muchtodo.local" http://172.19.0.2:32404/health



This confirmed:

- infrastructure working
- containers working
- storage working
- networking working
- services reachable
- Kubernetes deployment successful

…

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9. Security Considerations

- Non-root container user
- Environment-based secrets
- Isolated network (Kubernetes)

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10. Key DevOps Learnings

- Containerization is not enough → orchestration matters
- Kubernetes networking requires layered debugging
- Kind differs from cloud environments
- Ingress requires proper exposure strategy

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11. Final Status

- Docker build successful
- Docker Compose working
- Kubernetes deployment successful
- Ingress configured
- Application reachable via NodePort
- MongoDB integration confirmed

Author
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